Mimicking Biological Memory Consolidation in Spiking Neural Networks

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Abstract

Spiking Neural Networks (SNNs) offer a promising paradigm for energy-efficient neuromorphic computing, yet they pose significant challenges in maintaining knowledge across sequential tasks. In this work, we explore a biologically inspired framework for maintaining memory via neuronal consolidation. We investigate how specific neural patterns responsible for prediction can be identified in an SNN by introducing a **Potentiation-Factor (P-Factor)**, and propose a method to preserve the memory of past training while learning a new task. By tracking the potentiation of individual neurons based on local firing activity and reward signals, we implement a dual-phase learning method: (1) **Structural Locking** of identified expert neurons and (2) **Homeostatic Plasticity** for redundant novice neurons. Evaluated across the **Split-MNIST**, **N-MNIST**, and **5-Split Fashion-MNIST** benchmarks using both Class-Incremental (Class-IL) and Task-Incremental (Task-IL) evaluation protocols, this approach demonstrates that the P-factor successfully protects functional sub-networks, achieving strong retention on previous tasks while maintaining high plasticity for new learning. These results suggest that drawing inspiration from the structural dynamics of the brain offers a promising pathway toward continual learning in SNNs without the need for computationally expensive global curvature calculations or buffers of past data. To facilitate reproducibility, our source code is publicly available at https://github.com/vishesh-kumar-singh/Mimicking-Biological-Memory-Consolidation-in-SNNs.

Keywords: Spiking Neural Networks, Continual Learning, Neuromorphic Computing, Neuronal Consolidation, Catastrophic Forgetting

1. Introduction

The human brain possesses a remarkable ability to continuously learn new skills without erasing previously established memories. In biological systems, the organization and retention of these diverse memories over time is managed through a multi-stage process known as consolidation [2]. This capacity relies on the *stability-plasticity dilemma* [1]: neural circuits must remain plastic enough to encode novel information (learning) yet stable enough to preserve the traces of past experiences (memory). While some cognitive models suggest the brain utilizes internal replay to manage this balance [1], at the fundamental neuronal level, this is achieved not through global optimization, but through localized **Synaptic Consolidation**. In this work, we approximate this synapse-level process at the coarser granularity of individual neurons, trading strict biological fidelity for computational efficiency.

Neuroscience suggests that repeated activations of a neural pathway strengthen the connections between those neurons by triggering a cascade of chemical "tagging" processes (e.g., via CaMKII or PKMζ proteins) that structurally reinforce specific synapses [18, 19, 20], leading to a phenomenon known as Long-Term Potentiation. This is the primary mechanism responsible for memory retention in the biological brain. These reinforced neural pathways, known as engrams, lock in and prevent degradation of memory, leaving remaining the unassigned novice neurons free to learn new tasks.

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1.1. Spiking Neural Networks as a Biological Substrate

Spiking Neural Networks (SNNs) represent the "third generation" of neural models [3], designed to emulate the discrete, event-driven processing of the brain. Unlike traditional Artificial Neural Networks (ANNs) that rely on continuous activations, SNNs communicate via binary spikes, offering high energy efficiency suitable for neuromorphic hardware [4, 5].

However, standard SNN training algorithms (such as surrogate gradient descent [6]) treat all synaptic weights as equally plastic throughout the network's lifecycle. Because gradient descent indiscriminately updates all weights based solely on the new task's loss, the weights critical for previous tasks are inherently overwritten. This leads to the complete erasure of prior knowledge, a phenomenon known as Catastrophic Forgetting [7]. In terms of the Split-MNIST benchmark [8], this leads to almost 0.00% retention of prior knowledge.

1.2. The Gap: Biological Plausibility vs. Engineering Fixes

To mitigate forgetting, the field of Continual Learning (CL) has largely relied on engineering solutions that diverge from biological reality:

1. **Regularization (e.g., EWC [9]):** These methods constrain weight updates based on global loss curvature (Fisher Information). While mathematically rigorous, calculating second-order derivatives (Hessian) is computationally expensive, particularly over edge hardware like neuromorphic chips, and biologically implausible because synapses cannot access global network curvature.

2. **Replay Buffers [10]:** These methods store subsets of raw past data for rehearsal during new training. While effective, they require persistent memory overhead and do not reflect biological memory, which retains knowledge without explicit re-exposure to prior stimuli.
3. **Parameter Isolation (e.g., PackNet [12]):** These methods select and freeze a subset of network parameters after each task, allocating dedicated capacity per task. However, the selection criterion (e.g., weight magnitude) may not reflect functional importance in spiking networks.

1.3. Our Hypothesis: P-Factor Consolidation

In this work, we investigate a fundamental question: *Can simple, local Hebbian traces be sufficient to organize stable memory engrams in an SNN without global supervision?*

We propose a biologically inspired framework termed P-Factor Consolidation. Rather than treating memory consolidation as a global optimization problem, we model it as a local maturation process. We introduce a neuronal state variable, the Potentiation Factor (P-factor), which is loosely inspired by biological consolidation tags. This factor evolves via the local activity of neurons during training through the following mechanism:

- **Potentiation:** Neurons that repeatedly participate in successful firing events accumulate P, signaling their reliability. Scaling the effective weights of these neurons based on their P-factor promotes the independence of correctly responding circuits for predictions.
- **Engram Locking:** At task boundaries, neurons with high P-values are identified as the “Engram” (we borrow this term from neuroscience to denote the frozen functional sub-network that serves as the structural substrate of task memory) and structurally frozen.
- **Homeostatic Plasticity:** Low-confidence "Novice" neurons are left unlocked to naturally adapt and drift toward new features, providing fresh learning capacity without causing catastrophic network damage.

By simulating this dual-phase process of **Locking (Stability)** and **Natural Drift (Plasticity)**, we aim to demonstrate that bio-inspired, local plasticity rules can spontaneously segregate the network into “Expert” and “Novice” partitions. Our empirical results across Split-MNIST, event-based N-MNIST, and the dense 5-Split Fashion-MNIST sequence provide evidence that drawing inspiration from the structural dynamics of the brain is a promising and energy-efficient approach for continual learning in shallow SNN architectures.

2. Related Work

The challenge of Lifelong Learning (LL) has been extensively studied in Artificial Neural Networks (ANNs), but its translation to the energy-constrained, discrete domain of Spiking Neural Networks (SNNs) remains an active frontier. Existing approaches generally fall into three categories: Regularization, Replay, and Parameter Isolation.

2.1. Regularization and Generative Replay

Regularization approaches aim to penalize changes to important weights during subsequent tasks (e.g., EWC [9] and SI [8]). However, these methods often rely on high-precision gradient calculations during backpropagation, which are difficult to implement in SNNs where gradients are surrogate approximations [6]. Replay-based methods mitigate forgetting by interleaving old data with new training samples [10], or by training a separate generative network to "dream" pseudo-samples of past tasks [11]. While effective, this approach doubles the computational overhead and requires complex reconstruction losses that are non-trivial to implement in spike-based systems.

2.2. Parameter Isolation and the Continual Learning Taxonomy

In the broader continual learning taxonomy, methods are often split into dynamic architectural methods (growing capacity), memory-based rehearsal, and regularization. Recent work by Mirzadeh et al. [26] highlights that "architecture matters"; structural choices alone can rival purpose-built algorithms. Our P-Factor method sits at an interesting intersection: it is structural/architectural in mechanism (freezing neurons akin to PackNet [12]), but is driven purely by "neural" (local, activity-based) state variables rather than weight magnitudes.

This dual-memory framing, separating stable consolidated components from flexible plastic components, draws heavily on the complementary learning systems observed in mammalian brains, successfully abstracted in ANNs by Parisi et al. [27]. Unlike parameter isolation methods that rely on magnitude-based pruning (which wastes capacity on "silent but large" weights in SNNs), P-Factor tracks functional utility.

2.3. Recent SNN Architectures: Spiking CNNs and Transformers

Historically, SNN research has focused on shallow fully-connected architectures, but the field is rapidly scaling. Recent breakthroughs have demonstrated that deep Spiking CNNs (e.g., SEW-ResNet [36]) and Spike-driven Transformers [23, 24] can achieve state-of-the-art results on complex vision datasets [25]. For instance, Meta-SpikeFormer [21] and its scaled variants [22] optimize spike-driven self-attention for highly efficient representations.

These structural innovations highlight a critical distinction between *architectural memory* and *neural memory*. Spike-driven Transformers maintain memory structurally across the sequence via large activation buffers and complex routing. While powerful, this architectural memory often conflicts with the strict local routing of existing neuromorphic chips. Conversely, the P-Factor approach relies entirely on *neural memory*, defined as local, intrinsic state variables bound to individual neurons, which inherently respects the physical constraints of edge hardware.

2.4. Neuromorphic Adaptability and Closest Prior Art

The feasibility of deploying continual learning directly on neuromorphic hardware is gaining significant traction. Recent architectures like CLP-SNN [28], Genesis [29], and CLANE

[30] successfully implement event-driven spatiotemporal sparsity, activity-dependent metaplasticity, and self-normalizing local learning rules directly on-chip (e.g., Intel Loihi 2). These systems achieve orders-of-magnitude lower latency and energy consumption compared to edge-GPU baselines.

Our work is closely related to "neuron state dependent mechanisms" like NEO [31], which also advocates for a neuron-centric approach to continual learning. Furthermore, TACOS [32] and probabilistic metaplasticity approaches for memristor SNNs [33] leverage synaptic consolidation and metaplasticity without requiring explicit task-boundary awareness. Similarly, works by Allred & Roy [34] and Laborieux et al. [35] explore dopaminergic modulation and synaptic metaplasticity for stable lifelong learning. While these methods often operate at the $O(N^2)$ synaptic level, our P-Factor method intentionally approximates this at the $O(N)$ neuron level. By leveraging task-boundary signals, our method locks entire functional units, resulting in highly memory-efficient consolidation while avoiding the partial engram drift observed in fully plastic synapse-level methods.

2.5. Relation to Three-Factor Rules and Biological Models

Finally, our P-Factor mechanism aligns with the Three-Factor Learning Rule framework [16]. While algorithms like e-prop [17] solve temporal credit assignment for a single task, P-Factor acts as a complementary *meta-plasticity gate*, determining structural stability based on historical utility. Furthermore, while theoretical neuroscience models like the Cascade Model [13] and Synaptic Tagging and Capture (STC) [14, 15] focus on single-synapse dynamics, we scale this principle to network-level task retention, demonstrating that local consolidation rules can effectively support supervised continual learning in artificial SNN architectures.

3. Methodology

The core of our approach is the **Potentiation Factor (P)**, a local memory trace representing the structural stability and functional importance of a neuron. Unlike global importance metrics that require computing the entire loss landscape, our method relies on activity-dependent traces governed by local plasticity rules, making it highly efficient for neuromorphic implementations.

3.1. Neuron-Centric P-Factor: Bridging Biology and Engineering

In biological systems, synaptic tagging occurs at the level of individual dendritic spines (i.e., synapse-specific consolidation). However, implementing $O(N^2)$ synapse-specific variables in artificial SNNs is prohibitively memory-intensive for edge neuromorphic chips.

To address this, we diverge from strict biological realism and introduce a **Neuron-Centric** approximation. We track the reliability of the *post-synaptic neuron* as a proxy for the stability of its incoming weights. While biologically inspired by dendritic intrinsic excitability, we acknowledge this mechanism functionally acts as a form of *modular network engineering*. By selecting and freezing important neurons rather than individual synapses, the P-factor efficiently extracts task-specific functional sub-networks with only an $O(N)$ memory footprint.

Let $\mathbf{W} \in \mathbb{R}^{N_{out} \times N_{in}}$ be the weight matrix of the hidden layer, where the row vector $\mathbf{W}_{i,:}$ represents the incoming weights for neuron i. We introduce a state vector $\mathbf{P} \in \mathbb{R}^{N_{out}}$, initialized to zeros. The effective weights $\mathbf{W}_{eff}$ used during the forward pass are modulated by this factor:

$$\mathbf{W}_{eff}[i,:] = \mathbf{W}[i,:] \cdot (1 + P_i) \tag{1}$$

This is loosely analogous to the modulation of intrinsic excitability observed in biological neurons. By amplifying the effective synaptic strength of high-P neurons, we create a "winner-take-all" dynamic during training. This ensures that "Expert" neurons respond more robustly to familiar patterns, naturally segregating them from "Novice" neurons, which remain quieter and plastic.

3.2. Three-Factor Plasticity Rule

The evolution of P_i follows a **Three-Factor Rule** inspired by biologically plausible plasticity models: (1) Pre-synaptic activity (Input), (2) Post-synaptic firing rate (R_i), and (3) A global neuromodulatory signal (Reward or Error). This structure bridges the gap between local Hebbian traces and the supervised nature of the task.

3.2.1. Long-Term Potentiation (Reward)

Drawing analogy from biological Long-Term Potentiation (LTP), when the network correctly classifies an input, a global "Reward" signal increases the P-factor of active neurons to consolidate the successful circuit:

$$P_i \leftarrow P_i + \alpha_{LTP} \cdot R_i \cdot (P_{max} - P_i) \tag{2}$$

Here, R_i is the firing rate (spike count) of neuron i. The term $(P_{max} - P_i)$ acts as a soft saturation boundary (negative feedback), preventing unbounded growth. This ensures that only neurons consistently involved in correct predictions reach the maximum expert status ($P \approx P_{max}$).

3.2.2. Long-Term Depression (Punishment)

Analogously, if the network makes an **incorrect prediction**, a global "Punishment" signal triggers a depression of the P-factor, inspired by biological Long-Term Depression (LTD). Active neurons (which contributed to the noise/error) are penalized:

$$P_i \leftarrow \max(P_i - \alpha_{LTD} \cdot R_i, P_{min}) \tag{3}$$

We employ a linear decay for LTD to swiftly suppress erroneous neurons to P_{min}, effectively pruning unstable representations from the candidate engram.

Using Long-Term Potentiation and Long-Term Depression, we encourage the activity of selected correct neurons and discourage the activity of other neurons. This helps our engrams act independently of other nonessential neurons.

Algorithm 1 P-Factor Consolidation for Sequential Lifelong Learning

Input: Model $\mathcal{M}$, Task Sequence $\{\mathcal{D}_1, \ldots, \mathcal{D}_T\}$, Threshold τ
Initialize: $P_i \leftarrow 0$ for all neurons, $\text{Mask}_{expert} \leftarrow$ False
for task $t \in 1 \ldots T$ **do**
  Liberate: Reset P and output weights W_{out} for current task t classes
  // Phase 1: Train Current Task (New Learning)
  for epoch $e \in 1 \ldots E_t$ **do**
    for batch $(x, y) \in \mathcal{D}_t$ **do**
      Forward pass: $\hat{y} = \mathcal{M}(x)$
      Compute Loss with **Active Class Masking** (masking unseen/past classes)
      Compute Gradients ∇W via Backprop
      $\nabla W[\text{Mask}_{expert}] \leftarrow 0$ {Protect established engrams}
      Update W
      // Three-Factor Update
      if $\hat{y} == y$ **then**
        $P_i \leftarrow P_i + \alpha_{LTP} \cdot R_i \cdot (P_{max} - P_i)$ **for firing novice neurons**
      else
        $P_i \leftarrow \max(P_i - \alpha_{LTD} \cdot R_i, P_{min})$ **for firing novice neurons**
      end if
    end for
  end for
  // Phase 2: Offline Consolidation (Sleep)
  Calculate Threshold $\theta = \text{Percentile}(P, \tau)$
  $\text{Mask}_{new_expert} \leftarrow (P \geq \theta)$
  $\text{Mask}_{expert} \leftarrow \text{Mask}_{expert} \vee \text{Mask}_{new_expert}$ {Accumulate engram}
  Freeze: Lock hidden weights where Mask_{expert} is True
end for

3.3. Consolidation: Offline Reorganization ("Sleep")

At the boundary between tasks, we simulate a period of offline consolidation. The accumulated P-factors are used to structurally reorganize the network into stable and plastic partitions:

1. **Engram Identification:** We analyze the distribution of P values across the hidden layer. A percentile threshold τ (e.g., 80th percentile) serves as the boundary. Neurons with $P_i > \text{Percentile}(\mathbf{P}, \tau)$ are identified as the **Engram**. These engrams store the memory of the task the network was trained on.
2. **Synaptic Locking (Stability):** For every identified Engram neuron i, we freeze its **incoming synaptic weights** $\mathbf{W}_{i,:}$ (setting $\nabla \mathbf{W}_{i,:} = 0$). This ensures the neuron's feature selectivity remains invariant for future tasks and the memory is permanently locked in the neurons. Additionally, the output layer weights corresponding to Task A classes $(0 - 4)$ are frozen to preserve the classifier mapping.
3. **Homeostatic Plasticity (No-Reset):** Hidden neurons with low P values are deemed redundant. Instead of artificially re-initializing their synaptic weights, they are left unlocked to naturally adapt and drift toward features relevant for the new task. For the output layer, weights corresponding to Task B classes $(5 - 9)$ are reset to zero to facilitate new learning without classification bias from Task A.

4. Experimental Setup

4.1. Datasets and Task Formulation

We evaluate our proposed method on three continual learning benchmarks: Split-MNIST, its neuromorphic counterpart Split-N-MNIST, and 5-Split Fashion-MNIST. For both MNIST datasets, the 10 digit classes are partitioned into two disjoint tasks: Task A (digits 0-4) and Task B (digits 5-9). To test the limits of our consolidation mechanism in dense spatial environments, we formulate 5-Split Fashion-MNIST by dividing its 10 clothing classes into 5 sequential binary classification tasks. Unlike the sparse, easily separable digit features in MNIST, Fashion-MNIST images possess heavily overlapping spatial manifolds (e.g., shirts, coats, and pullovers), inducing severe attractor states that aggressively challenge structural plasticity. We employ a single-head evaluation setup, meaning the network is equipped with a single unified output layer of 10 neurons that must discriminate across all classes simultaneously. During evaluation, we report both Class-Incremental Learning (Class-IL) metrics (no task context) and Task-Incremental Learning (Task-IL) metrics (restricted to classes belonging to the specific task).

4.2. Network Architecture

Our experiments utilize a shallow, fully-connected Spiking Neural Network (SNN) topology. For Split-MNIST, the input dimension is 784, whereas for Split-N-MNIST, the spatio-temporal event streams are integrated into dense spike frames

of size $2 \times 34 \times 34 = 2312$. Both inputs project into a single hidden layer of 1024 Leaky Integrate-and-Fire (LIF) neurons, which then fully connects to the 10-neuron output head. The LIF neurons operate with a membrane time constant $\tau_m = 2.0$ and a firing threshold $V_{th} = 1.0$. Because the discrete nature of spike emission inherently precludes standard gradient-based optimization, we employ a fast sigmoid surrogate gradient function (with steepness $\beta = 10.0$) during the backward pass to approximate the non-differentiable step function of the spike threshold.

4.3. Implementation Details

All models are trained using the Adam optimizer with a base learning rate of 10^{-3} and a mini-batch size of 32. For the temporal dimension, both the static MNIST images (via Poisson rate encoding) and the N-MNIST event streams are simulated over $T = 100$ discrete timesteps per sample. Our proposed P-Factor consolidation mechanism relies on two dynamic scaling parameters: the Long-Term Potentiation rate ($\alpha_{LTP} = 0.01$) and the Long-Term Depression rate ($\alpha_{LTD} = 0.01$). The consolidation threshold τ, which determines the percentile of neurons locked as the Engram, is systematically evaluated across $\tau \in \{0.2, 0.4, 0.6, 0.8\}$ (representing freezing 20%, 40%, 60%, and 80% of the hidden neurons, respectively).

For the more challenging 5-Split Fashion-MNIST benchmark, the hyperparameters are adapted to accommodate the longer task sequence (4 consolidation boundaries instead of 1). Specifically, the depression rate is reduced to $\alpha_{LTD} = 0.005$ to prevent over-penalization of plastic neurons across successive tasks, and the consolidation threshold is set to $\tau = 0.15$ (freezing the top 15% of neurons per task boundary). This conservative freezing schedule ensures that approximately 40% of the hidden layer remains plastic by the onset of the final task, balancing structural stability with sufficient learning capacity.

Furthermore, to maintain gradient stability across extended task sequences, we implement **Active Class Masking** during training. The cross-entropy loss is computed exclusively over the classes that have been encountered up to the current task. This prevents freshly initialized, untrained output neurons from dominating the softmax denominator and skewing the gradients of established representations. Similarly, during the P-Factor update phase, future unseen classes are masked out to prevent them from dominating the argmax prediction, which would otherwise falsely trigger large Long-Term Depression (LTD) penalties on active plastic neurons.

4.4. Evaluation Metrics

To rigorously assess the trade-off between catastrophic forgetting and learning capacity, we measure performance using three primary metrics:

- **Task A Retention (Stability):** The classification accuracy on the Task A test set (digits 0-4) measured after the completion of Task B training. This quantifies how effectively the network mitigates catastrophic forgetting.
- **Task B Accuracy (Plasticity):** The final classification accuracy on the Task B test set (digits 5-9) at the end of training, indicating the network's capacity to absorb new information.
- **Combined Score:** The unweighted global accuracy across the entire test set (all 10 digits), serving as a holistic measure of the network's overall competency in solving both tasks simultaneously.

4.5. Training Protocol and Baselines

The training pipeline isolates tasks temporally. For Split-MNIST, Phase 1 (Maturation) trains the untrained network exclusively on Task A, while Phase 2 (Adaptation) exposes it solely to Task B, varied across $E \in \{1, 3, 5\}$ epochs per task. For the denser 5-Split Fashion-MNIST benchmark, we sequentially iterate through Tasks 1 to 5, testing severe temporal fragility by evaluating under $E \in \{1, 2, 3\}$ epochs per task.

To demonstrate the efficacy of our structural consolidation approach, we compare against several established continual learning baselines adapted for SNNs:

- **Fine-Tuning:** A naïve baseline where the network continuously trains on Task B without any regularization or consolidation.
- **Experience Replay (ER):** An oracle-like baseline that maintains a rehearsal buffer of 200 random samples from Task A, interleaved during Task B training.
- **Elastic Weight Consolidation (EWC) & Synaptic Intelligence (SI):** Two prominent gradient-based regularization methods that penalize changes to important weights, tested across multiple penalty strengths (λ, c).
- **PackNet & Random Freezing:** Structural baselines that freeze fixed percentages of weights based on magnitude pruning and random selection, respectively, providing a direct comparison to our P-Factor strategy.

5. Results and Discussion

All reported performance metrics represent the mathematical average across five independent experimental runs (using random seeds 42 through 46) to ensure statistical significance and reproducibility.

5.1. The Failure of Gradient-Based Regularization in SNNs

Traditional continual learning techniques fundamentally rely on the gradient of the loss with respect to the network parameters to estimate weight importance. However, as demonstrated in Figure 1, applying these gradient-based regularization methods, namely Elastic Weight Consolidation (EWC) and Synaptic Intelligence (SI), to Spiking Neural Networks yields severe performance degradation.

This systemic failure stems directly from the non-differentiable nature of spiking neurons. To enable backpropagation, SNNs must employ a surrogate gradient function that approximates the true derivative. EWC relies on calculating the diagonal of the Fisher Information Matrix (FIM), which requires accumulating

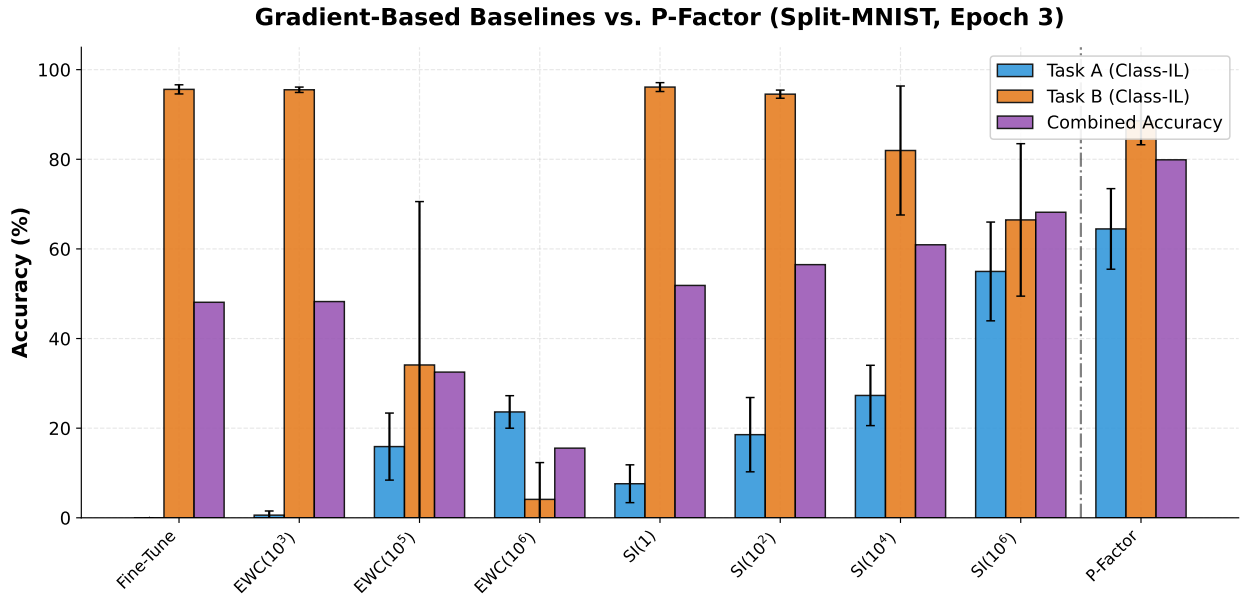


Figure 1: **Gradient-Based Baselines Performance Summary (Split-MNIST, Epoch 3).** The plot illustrates the severe trade-offs of gradient-based regularization in SNNs. Low penalties ($\lambda = 10^3$) fail to protect Task A, while extreme penalties ($c = 10^6$) severely bottleneck Task B learning. Our activity-driven P-Factor mechanism significantly outperforms all gradient-based regularization methods without sacrificing plasticity.

squared gradients. Because spike activity is inherently sparse and the surrogate gradient is merely a rough local approximation, the resulting FIM is intensely noisy and structurally sparse. Consequently, EWC fundamentally misidentifies the true "Engram" weights. As seen in Figure 1, low penalty constraints ($\lambda = 10^3$) fail to prevent catastrophic forgetting (Task A Class-IL Retention: 0.59%), whereas extreme constraints ($\lambda = 10^6$) arbitrarily lock the network, destroying its plasticity and plunging Task B Class-IL accuracy to 7.79%.

Similarly, Synaptic Intelligence (SI) attempts to track parameter importance by integrating the gradient trajectory over the entire training sequence. In an SNN, this means aggregating surrogate gradient approximation errors at every single timestep across thousands of iterations. This catastrophic noise accumulation renders the SI importance measure highly unreliable. While extreme SI constraints ($c = 10^6$) manage to artificially preserve some Task A Class-IL retention (54.97%), they do so at the expense of Task B learning capacity, which drops to 81.26% in Class-IL (compared to $> 97\%$ for the structural baselines). These results demonstrate that gradient-dependent consolidation is poorly suited for the discrete, non-differentiable optimization landscape of SNNs, motivating the need for our purely activity-driven P-Factor structural approach.

5.2. Engram Allocation: Utility vs. Magnitude

Parameter isolation methods typically select which weights to freeze based on their magnitude, assuming that large weights are the most important. To test this hypothesis in Spiking Neural Networks, we compare our utility-based P-Factor approach against PackNet (magnitude-based pruning) and a Random freezing control (Figure 2). Across all freezing thresholds ($\tau \in \{0.2, 0.4, 0.6, 0.8\}$), P-Factor consistently outperforms magnitude-based allocation. Most notably, at the strictest constraint ($\tau = 0.8$, meaning 80% of the network is frozen), P-Factor achieves a global Combined Class-IL accuracy of 79.88%, compared to PackNet's 65.86% and Random's 69.85%. This performance gap is driven primarily by Task A Class-IL retention, where the P-Factor engram (64.46%) effectively doubles the retention of PackNet (34.27%).

This significant disparity reinforces that magnitude is a flawed heuristic for SNNs. As discussed in Section 2.3, because spiking neurons employ a discontinuous thresholding mechanism, structurally freezing based on weight magnitude wastes valuable capacity on functionally silent neurons. By tracking the P-Factor, a direct measure of historical spiking activity and correct prediction involvement, our method ensures that the protected engram consists of functionally active circuits rather than just statistically large parameters.

The competitive performance of Random Freezing is mathematically explained by the inherent sparsity of SNN activations. To provide direct empirical evidence for this, we analyzed the full firing-rate distribution of the 1024 hidden neurons over the Task A validation set (Figure 3). The distribution exhibits a steep long-tail sparsity curve. We quantitatively define the "active, task-relevant" neurons as the top 20% most frequently firing neurons ($n = 204$). At our strictest capacity budget ($\tau = 0.8$, meaning 819 neurons are frozen), a Random Freezing strategy captures these critical neurons with a coverage of exactly 80.08%, strictly following statistical probability. In contrast, our P-Factor mechanism successfully identifies and protects 100.00% of these active functional circuits within the same 819-neuron budget. This near-perfect coverage of task-relevant neurons explains the ~10% retention performance gap between P-Factor and Random Freezing.

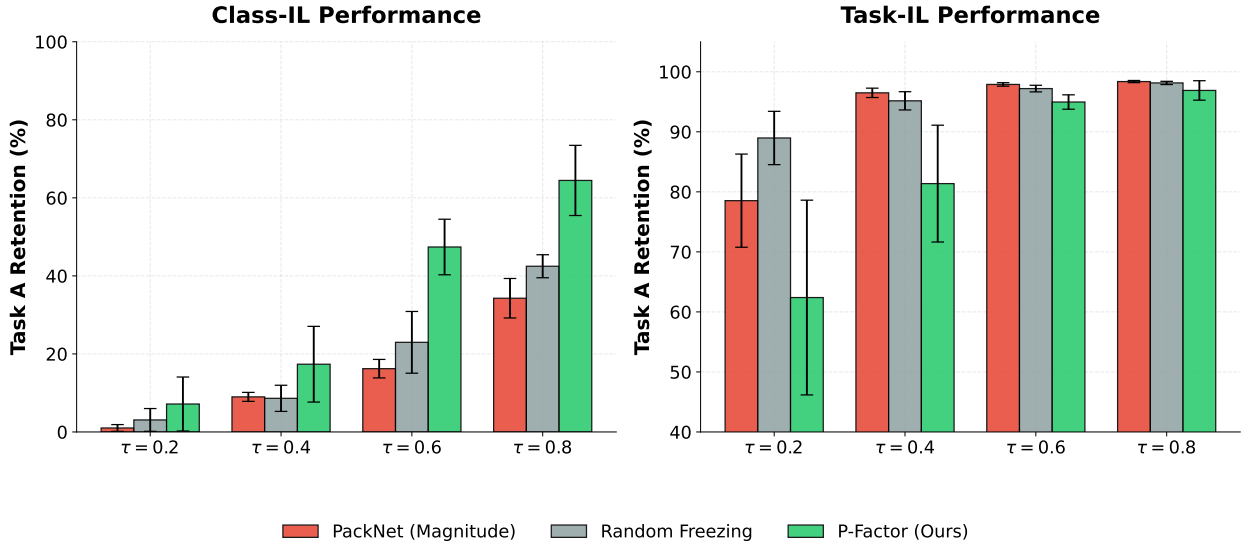


Figure 2: **Comparison of Structural Engram Allocation Strategies (Split-MNIST, Epoch 3).** The multi-plot demonstrates the superiority of the P-Factor mechanism over magnitude-based pruning (PackNet) and Random freezing across multiple capacity constraints ($\tau \in \{0.2, 0.4, 0.6, 0.8\}$). The left subplot shows Class-IL retention (measuring the preservation of the inter-task decision boundary), while the right subplot shows Task-IL retention (measuring the absolute preservation of internal representation extractors). P-Factor consistently achieves the highest retention.

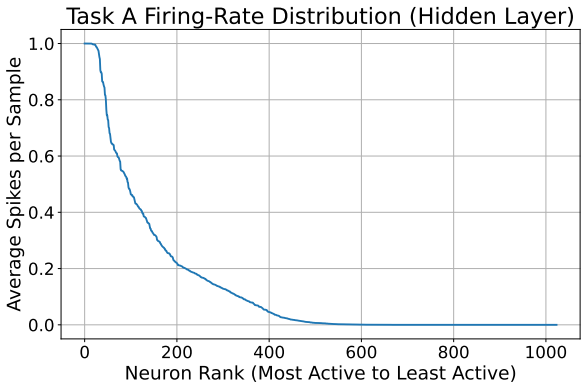


Figure 3: **Task A Firing-Rate Distribution.** The average spikes per sample for all 1024 hidden neurons over the Task A validation set. The distribution exhibits a steep long-tail sparsity curve. The top 20% most active neurons are defined as 'task-relevant'. P-Factor covers 100.00% of these critical neurons at $\tau = 0.8$, whereas Random Freezing covers only 80.08%.

5.3. Longitudinal Dynamics: Task A Retention Trajectory

To understand how the duration of adaptation impacts the P-Factor consolidation process, we plotted the network's Task A retention trajectory (Class-IL vs. Task-IL) across 1 to 5 epochs per task at the highest capacity constraint ($\tau = 0.8$) (Figure 4). This specific trajectory reveals two key insights into the behavior of a fully protected structural engram:

1. Absolute Structural Protection (Task-IL): Given sufficient engram capacity ($\tau = 0.8$), the Task-IL accuracy remains exceptionally high and stable ($\approx 94 - 96\%$) across all adaptation durations. Because Task-IL evaluation ignores output head biases between tasks, this stability confirms that the P-Factor mechanism successfully isolates and physically protects the core feature extractors. As long as the capacity threshold is high enough, the frozen engram prevents weight degradation, allowing the network to reliably retain the internal representations necessary to classify Task A digits.

2. The Impact of Plasticity Creep (Class-IL): While the internal representations are heavily protected at this threshold, the Class-IL accuracy (which tests the network's ability to choose between Task A and Task B outputs) remains highly sensitive to the duration of Task B training. Class-IL retention peaks at 64.46% at Epoch 3. However, extending training to Epochs 4 and 5 triggers a decline in Class-IL accuracy. This occurs because excessive training iterations on Task B cause the plastic novice neurons to over-specialize and artificially inflate their output magnitude, skewing the overall decision boundary away from the frozen Task A engram. This demonstrates that even with a highly protected internal structure, SNN consolidation operates within a strict temporal window where excessive rehearsal paradoxically degrades the balanced multi-task representation at the output layer.

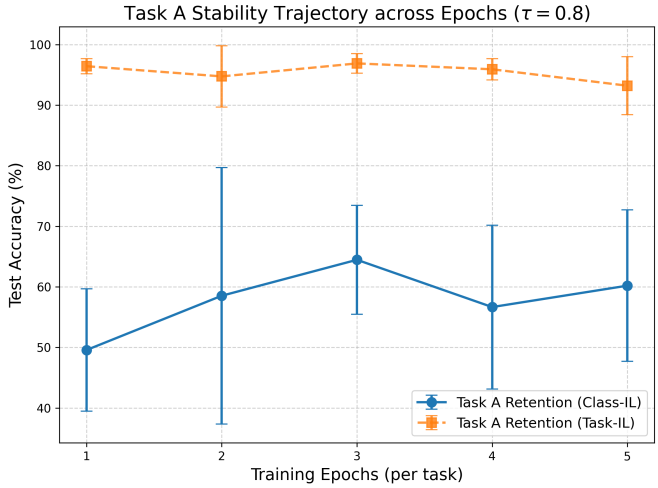


Figure 4: Task A retention trajectories (Class-IL vs. Task-IL) across Epochs 1 to 5 for $\tau = 0.8$. Standard deviation is shaded/indicated by error bars.

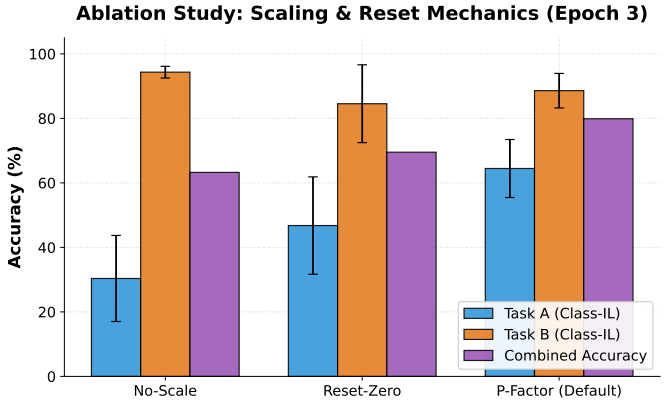


Figure 5: **Ablation Study: Forward Pass Scaling and Reset Mechanics (Split-MNIST, Epoch 3).** The plot demonstrates the functional necessity of scaling the forward pass by the P-Factor (improving combined accuracy from 63.25% to 79.88%) and validates that leaving unselected neurons intact (No-Reset) outperforms forcing artificial neurogenesis (Reset-Zero).

5.4. Ablation Studies: Mechanics of the Engram

To isolate the contributions of the individual mechanisms within our consolidation framework, we conducted ablation studies at the $\tau = 0.8$ threshold after 3 epochs of adaptation (Figure 5).

Forward Pass Scaling. In the default P-Factor method, we modulate the effective synaptic weights during the forward pass based on their potentiation ($\mathbf{W}_{eff} = \mathbf{W} \cdot (1+P)$). When we ablate this scaling (the "No-Scale" variant), the combined accuracy drops significantly from 79.88% to 63.25%, driven primarily by a collapse in Task A Class-IL retention (from 64.46% down to 30.36%). This confirms that amplifying the influence of expert neurons during training is crucial; it creates a "winner-take-all" dynamic that forces the network to rely on the established engram for familiar inputs, preventing novice neurons from interfering with prior knowledge.

Novice Reset Strategy. At the task boundary, redundant novice neurons are left unlocked to learn the new task. We compared our default "No-Reset" strategy (leaving their weights intact) against a "Reset-Zero" baseline where all plastic incoming weights are strictly re-initialized to zero. As shown in Figure 5, forcing a hard reset severely degrades performance (combined accuracy drops to 69.51%). This suggests that the untrained novice neurons still retain weak, generalizable feature representations from Phase 1. Allowing them to naturally drift ("No-Reset") provides a warmer start for Task B without harming Task A, demonstrating that homeostatic plasticity is more effective than artificial neurogenesis in this architecture.

5.5. Scaling to Dense Feature Spaces: 5-Split Fashion-MNIST

To stress-test the P-Factor consolidation mechanism beyond the sparse, easily separable digit manifolds of MNIST, we evaluate on 5-Split Fashion-MNIST, where the 10 clothing categories are divided into 5 sequential binary classification tasks. This setting is substantially more challenging for three reasons: (1) Fashion-MNIST features exhibit heavy spatial overlap (e.g., T-shirts vs. Pullovers share nearly identical silhouettes), (2) the network must consolidate memory across 4 sequential boundaries (vs. 1 for Split-MNIST), and (3) the available plastic capacity diminishes with each successive task.

Figure 6 presents the full progression trajectory for the P-Factor method at 3 epochs per task. The trajectory reveals several critical dynamics:

Early-Task Retention. At the Task-IL level, the P-Factor mechanism maintains strong retention for Tasks 1-3 throughout the entire sequence. After all 5 tasks, Task 1 still achieves 83.5% ± 19.0 and Task 3 achieves 94.5% ± 10.8 in Task-IL, demonstrating that the structural engrams formed during early consolidation remain functionally intact despite subsequent weight updates.

Capacity Saturation and Learning Collapse. A stark failure mode emerges at Tasks 4 and 5. With $\tau = 0.15$ frozen per boundary, approximately $3 \times 15\% = 45\%$ of the hidden layer is structurally locked by the onset of Task 4, rising to $4 \times 15\% = 60\%$ by the onset of Task 5 and leaving only ~40% of neurons plastic for the final task. The combination of (a) constrained plastic capacity and (b) the accumulated P-factor scaling on frozen neurons produces exploding effective weights in the hidden layer. When these amplified activations project into the freshly initialized Task 4/5 output heads, the resulting gradient magnitudes destabilize learning. This manifests as collapse to near-random accuracy (Task 4 Class-IL: 9.1% ± 18.1; Task 5 Class-IL: 18.7% ± 37.4).

High Variance as Evidence of Structural Brittleness. The exceptionally large standard deviations (~20-40%) on Tasks 4 and 5 are not statistical noise but reflect a **high-variance, seed-dependent** outcome. In particular, seed 44 discovers a separable manifold configuration where the remaining plastic neurons successfully specialize (Task 5 Class-IL: 93.5%), while other seeds collapse (0.0%). This variance confirms that the method operates at a critical capacity threshold: when the stochastic initialization of the remaining plastic neurons happens to align with the task's feature subspace, learning succeeds; otherwise, the exploding gradient dynamics prevent convergence.

Comparison with Baselines. The full progression matri-

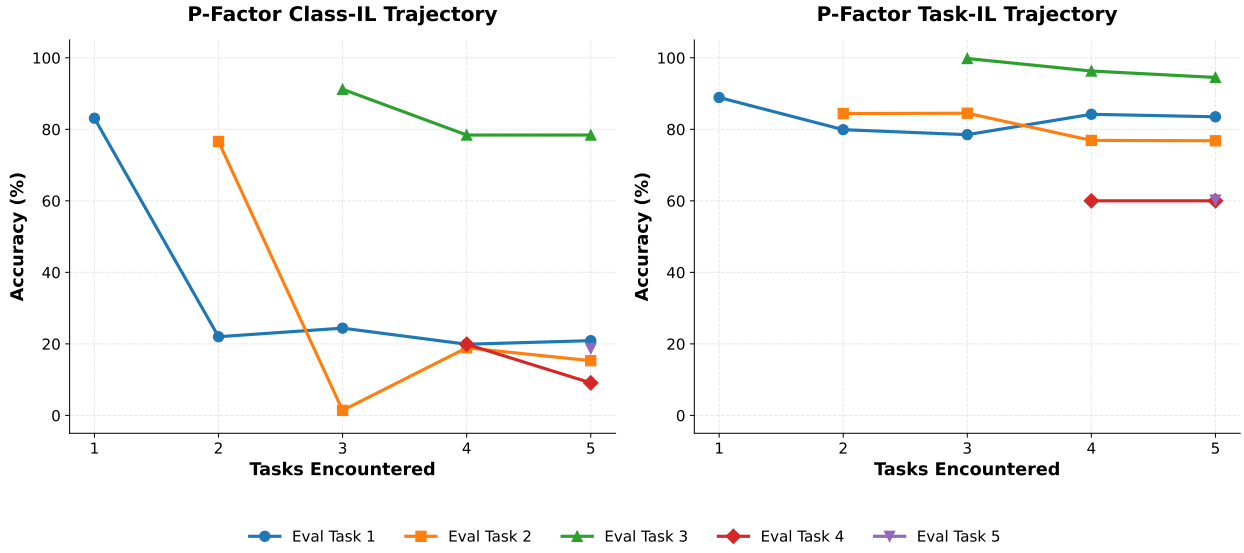


Figure 6: **5-Split Fashion-MNIST (Epoch 3): P-Factor Progression Over Time.** The trajectory reveals strong early-task retention but capacity-driven collapse on Tasks 4-5. Task-IL retention (right) remains high for early tasks, while Class-IL (left) suffers due to output bias.

ces for Experience Replay, Fine-Tuning, and gradient-based baselines (EWC, SI) are provided in Appendix Appendix B. As observed in simpler benchmarks, gradient-based regularization completely fails to maintain multi-task boundaries in SNNs. Even at their optimal constraints, Elastic Weight Consolidation ($\lambda = 10^5$) and Synaptic Intelligence ($c = 10^6$) suffer from severe catastrophic forgetting on early tasks (Task 1 final Class-IL of 8.8% and 15.6%, respectively) due to the cumulative noise of surrogate gradients across 5 task phases. While ER achieves higher absolute Class-IL accuracy (final average 63.2% vs. 28.5% for P-Factor), this advantage is driven by the replay buffer re-consolidating early tasks during later training, an external memory mechanism that the P-Factor explicitly avoids. Crucially, ER also suffers from **Surrogate Gradient Starvation** in this dense feature environment: the continuous replay of Task 1 data drives the network into an attractor state where dominant Task 1 neurons strongly inhibit late-task neurons. Because the surrogate gradient function evaluates to zero for neurons whose membrane potential remains far below the firing threshold, the learning signal for new tasks vanishes. Fine-Tuning, as expected, exhibits catastrophic forgetting (Task 1 final Class-IL: 0.0%).

Comparison with Structural Baselines. While gradient-based methods fail due to surrogate noise, structural baselines (PackNet and Random Freezing) highlight the weakness of naive freezing heuristics in dense environments. As shown in Appendix Appendix B, both PackNet and Random Freezing suffer a complete collapse in Class-IL accuracy for early tasks (Task 1 final Class-IL drops to 0.0%). Although their Task-IL retention remains high (indicating the physical weights are preserved), their inability to select functionally relevant "expert" neurons causes their output layer decision boundaries to be completely overwritten by subsequent tasks. In contrast, the P-Factor's utility-driven allocation retains a functioning decision boundary (Task 1 final Class-IL: 20.9%), proving that activity-based engrams are essential for multi-task separation in overlapping feature spaces.

These results demonstrate that while P-Factor consolidation strongly mitigates forgetting for early tasks via structural isolation, it imposes a hard architectural limit on the maximum task sequence length. Overcoming this ceiling requires either active capacity expansion (neurogenesis), dynamic threshold scheduling, or hybrid approaches that combine structural isolation with lightweight replay.

5.6. Scaling to High-Entropy Temporal Streams (N-MNIST)

To evaluate the robustness of our consolidation mechanism on high-entropy spatio-temporal data, we extend our analysis to the neuromorphic N-MNIST dataset. Unlike static images, N-MNIST features asynchronous event streams with severe temporal noise, which aggressively challenges structural plasticity.

Before evaluating structural isolation, we examine the performance of non-structural baselines on the dense event streams of N-MNIST (see Appendix B.10 for full quantitative matrices). As expected, **Fine-Tuning** suffers from complete catastrophic forgetting, dropping to 0.00% Class-IL accuracy on Task A. While **Experience Replay (ER)** perfectly preserves the memory (91.95% Class-IL) by continuously rehearsing the noisy spatio-temporal patterns, it requires storing and replaying raw data, which our method avoids.

Crucially, the gradient-based regularization methods fail severely, further reinforcing the unreliability of surrogate gradients. **Elastic Weight Consolidation (EWC)** manages a mere 6.32% Class-IL retention even at extreme constraints ($\lambda = 10^6$). The severe temporal noise of N-MNIST causes the Fisher Information Matrix (computed via noisy surrogate gradients over $T = 100$ timesteps) to completely misidentify critical synapses. **Synaptic Intelligence (SI)** performs comparatively better than EWC under extreme constraints ($c = 10^6$), reaching 34.74% Class-IL retention. However, this is still significantly worse than

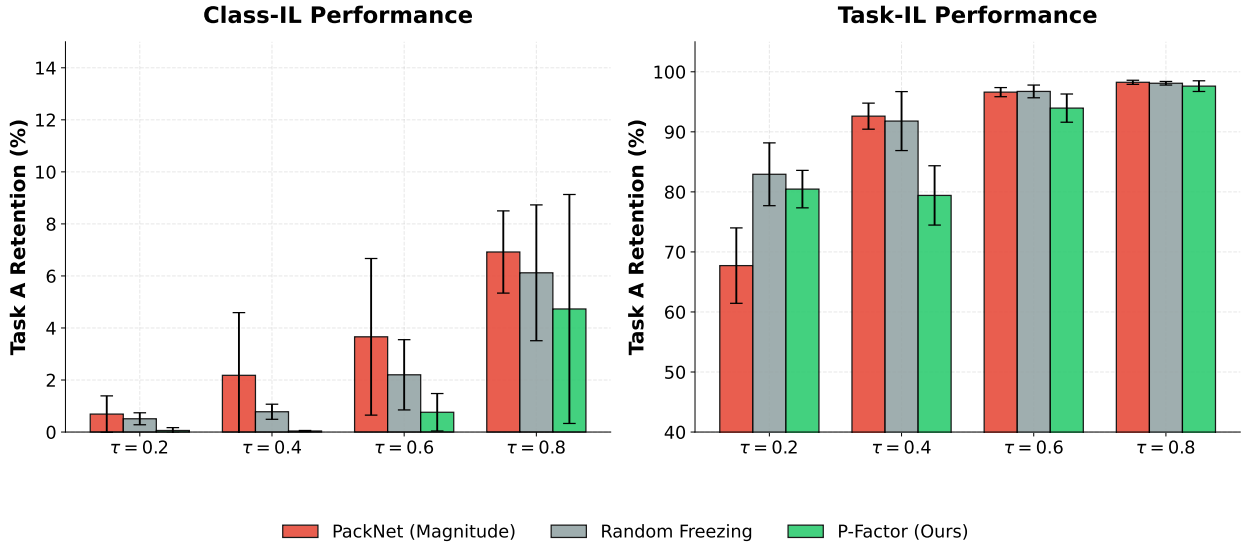


Figure 7: **Comparison of Structural Engram Allocation Strategies (Split-N-MNIST, Epoch 3).** The plot illustrates the severe task-recency bias present in high-entropy spatio-temporal streams. While all structural methods experience a near-total collapse in Class-IL retention (left) due to unconstrained output layer drift, their Task-IL retention (right) remains highly robust (> 95% at high capacities). The P-Factor mechanism achieves competitive Task-IL retention through its functionally-driven engram, confirming that activity-based tagging protects core feature extractors even under severe temporal noise.

replay and requires aggressively over-regularizing the network to force stability.

Transitioning to structural isolation methods, we observe that while the unconstrained output layer suffers from task-recency bias (causing Class-IL to collapse for most methods), the internal engrams physically survive. Figure 7 illustrates the structural comparison across varying capacity constraints ($\tau \in \{0.2, 0.4, 0.6, 0.8\}$).

Across all evaluated structural isolation methods, the performance diverges significantly based on the available structural capacity. At strict capacity constraints ($\tau = 0.2$, freezing only 20% of the hidden layer), magnitude-based PackNet fails to identify the critical features, dropping to 67.72% Task-IL retention for Task A, whereas the P-Factor maintains a more robust 80.46%. However, an important failure mode emerges as the consolidation threshold increases. At higher capacities ($\tau \in \{0.4, 0.6, 0.8\}$), the P-Factor is consistently beaten by both PackNet and Random Freezing in both Task-IL and Class-IL retention metrics. For example, at $\tau = 0.4$, Random Freezing achieves 91.79% Task-IL, while the P-Factor lags at 79.41%.

This inversion reveals a fundamental vulnerability of activity-driven consolidation in high-entropy environments. Because N-MNIST features severe temporal noise, neurons with the highest firing utility (the highest P-Factors) during Task A often include those capturing high-frequency background noise rather than stable semantic features. When the P-Factor mechanism freezes a large population ($\geq 40\%$) of these noise-driven "experts" and continuously amplifies their forward-pass activations during Task B, they project massive, chaotic signals into the unconstrained output layer. This exacerbates output-layer gradient explosion, accelerating Class-IL collapse and degrading Task-IL retention compared to the uniform, unscaled distribution of Random Freezing. Thus, while the P-Factor protects core features under tight biological constraints, its reliance on firing statistics makes it uniquely susceptible to noise-induced capacity saturation in chaotic event streams.

5.7. Hyperparameter Sensitivity Analysis

To systematically understand the method's reliance on specific parameter bounds, we conducted a hyperparameter sensitivity sweep on the Split-MNIST benchmark. The core mechanism of P-Factor consolidation hinges on the delicate equilibrium between Long-Term Potentiation (α_{LTP}) and Long-Term Depression (α_{LTD}). As shown in the heatmaps (Figure 8), this balance dictates the network's traversal across the stability-plasticity continuum.

Analyzing the empirical data reveals three distinct operational regimes governed by these hyperparameters:

1. **Optimal Balance (The Sweet Spot):** When both potentiation and depression rates are suitably high and balanced (e.g., $\alpha_{LTP} = 0.01, \alpha_{LTD} = 0.015$), the network successfully navigates the stability-plasticity dilemma. In this regime, the network achieves its peak Combined Class-IL accuracy (> 81%). It preserves a highly functional Task A engram (Task A Class-IL ≈ 68.6%) while retaining more than enough unconstrained novice neurons to master the new task (Task B Class-IL ≈ 93.7%). The simultaneous presence of strong LTP (to securely lock expert neurons) and strong LTD (to rapidly penalize noisy neurons) ensures that the formed engram is compact and highly discriminative.
2. **Stability Bias (Plasticity Collapse):** In parameter configurations where depression heavily outweighs potentiation at a moderate scale (e.g., $\alpha_{LTP} = 0.005, \alpha_{LTD} = 0.015$), the network exhibits an extreme bias toward prior knowledge. Task A retention soars to ≈ 88.4%, but Task B

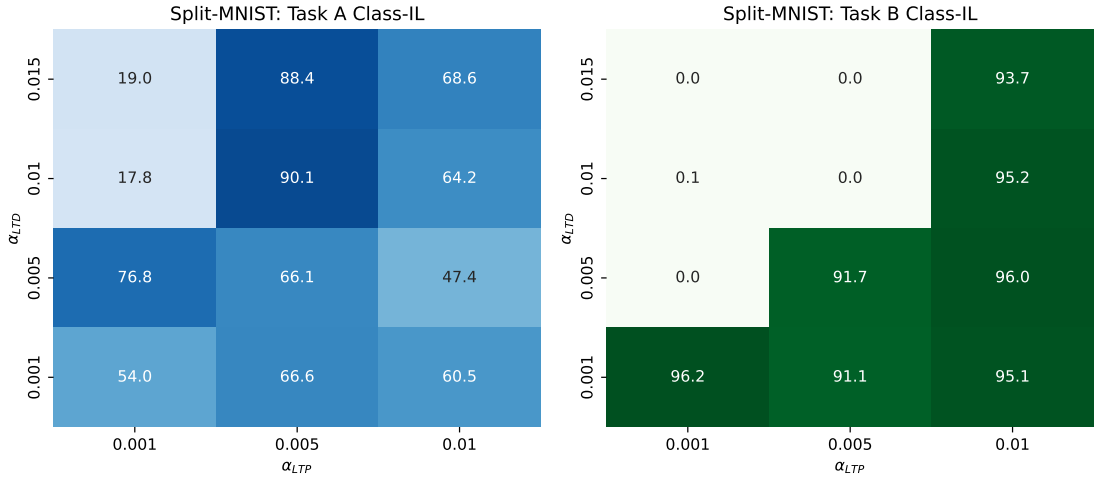


Figure 8: **Hyperparameter Sensitivity on Split-MNIST (Epoch 3, $\tau = 0.8$).** The heatmaps illustrate the stability-plasticity dynamics. With an optimal balance of parameters ($\alpha_{LTP} = 0.01, \alpha_{LTD} = 0.015$), the network successfully mitigates forgetting (Task A Class-IL $\approx 68.6\%$) while maintaining high plasticity (Task B Class-IL $\approx 93.7\%$).

learning completely collapses (0.0%). In this state, the hyper-penalization during Phase 1 likely forces the network into an attractor state where too much of the hidden layer is functionally entangled or frozen, leaving insufficient plastic capacity to map the novel Task B features.

3. **Representational Collapse:** Conversely, when both parameters are excessively low (e.g., $\alpha_{LTP} = 0.001, \alpha_{LTD} = 0.015$), the localized memory traces fail to accumulate effectively. The network struggles to differentiate expert neurons from novice neurons, resulting in a near-total failure to learn or retain either task (Task A $\approx 19.0\%$, Task B $\approx 0.0\%$).

These distinct regimes empirically validate our hypothesis that local, activity-driven traces (analogous to Hebbian mechanisms) are sufficient to organize stable functional sub-networks in an SNN without global supervision, provided that the dynamics of neuronal potentiation and depression are properly balanced.

6. Energy and Computational Cost Analysis

A major advantage of the P-Factor consolidation mechanism is its computational and memory efficiency, particularly when deployed on neuromorphic hardware. Unlike regularization methods like EWC or Synaptic Intelligence (SI) that must store dense parameter-scale matrices of size $O(W)$ (effectively doubling or tripling the ~ 3.1 MB parameter footprint), the P-Factor tracks utility at the neuronal level. This requires storing only a single scalar per hidden neuron ($O(N)$). For our 1024-neuron hidden layer, this adds a negligible ~ 4 KB of memory overhead.

Furthermore, our energy analysis (using Synaptic Operations, or SynOps, as a proxy for dynamic energy consumption) reveals that the P-Factor maintains high activation sparsity ($\sim 86\%$ silent neurons per timestep, measured empirically by averaging the fraction of non-firing neurons across all timesteps and samples during Task A evaluation), resulting in approximately 10.6×10^6 Inference SynOps per sample on the Split-MNIST benchmark. Because all baselines share the identical underlying architecture and firing thresholds, their forward-pass inference SynOps are virtually identical ($\sim 10.6 \times 10^6$). The true efficiency gains of the P-Factor emerge during training: crucially, because frozen neurons (e.g., 80% of the layer at $\tau = 0.8$) are strictly excluded from gradient computation and weight updates, the backward pass energy cost drops significantly compared to Fine-Tuning. In contrast, Experience Replay (ER) inherently increases energy costs by continually interleaving past samples into the training stream. Table 1 summarizes these efficiency gains.

6.1. *Quantifying Collateral Locking: Neuron vs. Synapse Level Consolidation*

While true biological consolidation, such as Synaptic Tagging and Capture [14], operates at the resolution of individual synapses, our P-Factor method serves as a neuron-level engineering approximation. A critical limitation of any neuron-level method is *collateral locking*: when a neuron is frozen to preserve its functional utility, all of its incoming synapses are simultaneously locked, regardless of their individual magnitudes. To quantify this structural waste, we analyzed the weight distribution of the frozen engram at our strictest constraint ($\tau = 0.8$). We found that approximately 23.5% of the frozen synapses possessed near-zero magnitudes ($|w| < 0.01$). These synapses are functionally useless for Task A but are collateral casualties of the neuron-level freeze, reducing the network's available plasticity for Task B.

To evaluate whether this collateral locking is an acceptable trade-off, we implemented an "Ideal Synaptic-Level P-Factor" baseline, where potentiation and locking are applied independently to all 1024×784 incoming synapses. As shown in Table 2, while the ideal synaptic method avoids collateral locking and maintains high plasticity for Task B (93.96% accuracy), it un-

Table 1: Computational Overhead and Energy Efficiency Comparison

Method	Memory Overhead	Relative Energy Cost	Inference SynOps/Sample	Complexity
Fine-Tuning	None	1.0× (Baseline)	$\sim 10.6 \times 10^6$	$O(1)$
ER (Buffer=200)	$O(\|B\| \times D)$ ($\sim$ 156 KB)	$> 1.0\times$ (Replay passes)	$\sim 10.6 \times 10^6$	$O(\|B\|)$
EWC	$O(W)$ ($\sim$ 6.2 MB)	1.0×	$\sim 10.6 \times 10^6$	$O(W)$
SI	$O(W)$ ($\sim$ 6.2 MB)	1.0×	$\sim 10.6 \times 10^6$	$O(W)$
P-Factor (Ours)	$O(\mathbf{N})$ ($\sim$ 4 **KB**)	$< \mathbf{1.0}\times$ **(Sparse backward pass)**	$\sim \mathbf{10.6 \times 10^6}$	$O(\mathbf{N})$

Method	Retention (Task A Class-IL)	Plasticity (Task B Class-IL)	Memory Overhead
Ideal Synaptic P-Factor	53.99%	93.96%	$\sim$ 3 MB
Neuron-Level P-Factor	64.46%	95.30%	**4 KB**

Table 2: **Cost of Collateral Locking.** Comparison between true synaptic consolidation ($O(N^2)$) and our neuron-level engineering approximation ($O(N)$) at $\tau = 0.8$.

expectedly degrades Task A retention (53.99%) compared to the neuron-level method (64.46%). This reveals a crucial network dynamic: leaving non-critical synapses plastic within an engram allows subsequent tasks to induce partial synaptic drift, gradually eroding the engram's original functional response. By rigidly locking entire neurons, collateral locking paradoxically *improves* stability.

Furthermore, the synaptic method requires tracking a state variable for every synapse ($O(N^2)$ memory), resulting in $\sim$ 3.2 MB of memory overhead, compared to just 4 KB for the neuron-level approximation ($O(N)$ memory). Given that neuromorphic hardware typically features severely constrained per-core SRAM, this 800× reduction in memory overhead, combined with superior retention, makes the neuron-level approach a highly favorable engineering choice.

7. Limitations and Future Work: Scaling to Deeper Architectures

While our P-Factor mechanism successfully demonstrates energy-efficient, local memory consolidation on fully-connected SNNs and event-based sensory streams (e.g., N-MNIST), it is important to acknowledge the current architectural limitations. Extending this methodology to deeper, spatially complex architectures like Convolutional Spiking Neural Networks (CSNNs) presents unique challenges. In convolutional layers, weights are shared across spatial dimensions, meaning a single feature map acts on multiple local receptive fields. Applying a neuron-centric P-Factor would require either (a) consolidating entire feature maps homogeneously or (b) adapting the mechanism to tag specific spatial filters. However, because our P-Factor strategy operates with an $O(N)$ memory footprint (tracking stability per neuron rather than per synapse), it is theoretically highly scalable to these deeper architectures compared to traditional $O(N^2)$ synaptic tagging approaches. Future work will focus on integrating P-Factor consolidation with multi-layer CSNNs to tackle large-scale visual recognition tasks such as CIFAR-10 and ImageNet within neuromorphic constraints.

8. Conclusion

In this work, we presented P-Factor Consolidation, a novel framework inspired by biological neuronal consolidation for practical continual learning in Spiking Neural Networks. By introducing a local, activity-dependent state variable, the Potentiation Factor, we demonstrated that the complex phenomenon of memory consolidation can be effectively approximated without the need for computationally expensive global curvature estimates (e.g., Hessian matrices) or external replay buffers.

Our experiments across three benchmarks of increasing difficulty revealed several key findings. On Split-MNIST, the P-Factor mechanism achieved a combined accuracy of 79.88% at $\tau = 0.8$, significantly outperforming both magnitude-based (PackNet: 65.86%) and random (69.85%) freezing strategies, confirming that activity-driven engram allocation is more robust than weight-magnitude heuristics in the discrete optimization landscape of SNNs. On the neuromorphic N-MNIST dataset, we showed that the internal engram physically survives severe temporal noise (Task-IL retention > 97%), even when the unconstrained output layer suffers from task-recency bias. The 5-Split Fashion-MNIST experiment demonstrated strong early-task retention across 5 sequential boundaries, while simultaneously exposing a hard architectural capacity limit: with ~60% of neurons frozen by Task 4, the remaining plastic capacity becomes insufficient for dense, overlapping feature spaces.

These results highlight both the promise and the current limitations of structural consolidation in shallow fully-connected SNNs. The P-Factor mechanism operates with an $O(N)$ memory footprint, requires no gradient-based importance estimation, and naturally reduces backward-pass energy costs. These properties are highly amenable to deployment on neuromorphic hardware. However, the observed capacity saturation on longer task sequences and the sensitivity to hyperparameter selection underscore the need for future work on dynamic threshold scheduling, active capacity expansion (neurogenesis), and extension to deeper convolutional and recurrent SNN architectures.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.

During the preparation of this work, the author used Gemini in order to refine the content of the manuscript. After using this, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Appendix A. Network Architecture and Experimental Hyperparameters

All experiments in this study employ a 2-layer Fully-Connected Spiking Neural Network (SNN) utilizing Leaky Integrate-and-Fire (LIF) neurons. The network capacity is kept constant across all benchmarks to ensure a fair evaluation of architectural limits under catastrophic interference.

- **Input Dimension:** 784 (Split-MNIST and 5-Split Fashion-MNIST), 2312 (N-MNIST).
- **Hidden Layer:** 1024 LIF neurons.
- **Output Layer:** 10 LIF neurons.
- **Neuron Model Parameters:** Membrane time constant $\tau_m = 2.0$ (corresponding to a leak factor $\alpha = e^{-1/\tau_m} \approx 0.61$), spiking threshold $V_{th} = 1.0$.
- **Surrogate Gradient:** FastSigmoid function with steepness parameter $\beta = 10.0$.
- **Optimization:** Adam optimizer with a constant learning rate of 1×10^{-3}.
- **Batch Size:** 32.
- **P-Factor Consolidation Rates:** $\alpha_{LTP} = 0.01$, $\alpha_{LTD} = 0.01$ for MNIST/N-MNIST, and $\alpha_{LTD} = 0.005$ for Fashion-MNIST (optimized via hyperparameter sweeps on the validation set).
- **Reproducibility:** All reported metrics are averaged across 5 independent random seeds $(42, 43, 44, 45, 46)$ to ensure statistical significance.

Appendix B. Detailed Experimental Results

For completeness, we provide the full breakdown of experimental results across all training durations ($E \in \{1, 3, 5\}$) and consolidation thresholds ($\tau \in \{0.2, 0.4, 0.6, 0.8\}$). This longitudinal view is critical for understanding the temporal fragility of SNNs and the onset of plasticity creep during extended adaptation.

Table B.3: **Epoch 1: P-Factor vs Baselines (Mean ± Std).** The following tables summarize the performance of the network after a single epoch of training on Task B. At this early stage, the network has not yet fully overwritten its synaptic weights, leading to artificial retention across most structural baselines. As observed, simply freezing 80% of the neurons randomly yields a competitive retention on Task A. This suggests that during brief adaptation phases, catastrophic forgetting is partially mitigated by the network's inherent inertia rather than the efficacy of the consolidation algorithm itself.

Method	**Task A (Class-IL)**	**Task A (Task-IL)**	**Task B (Class-IL)**	**Task B (Task-IL)**	**Combined**
Fine-Tuning (Baseline)	0.04% ± 0.07	19.27% ± 0.30	95.59% ± 1.51	95.60% ± 1.52	46.85%
ER (Buffer=200)	89.46% ± 2.58	95.22% ± 1.95	95.29% ± 0.79	96.61% ± 0.62	92.65%
EWC ($\lambda = 10^3$)	2.13% ± 3.57	31.59% ± 9.51	96.15% ± 0.64	96.15% ± 0.64	48.40%
EWC ($\lambda = 10^5$)	9.68% ± 11.85	25.02% ± 3.43	79.75% ± 19.81	80.46% ± 18.94	44.21%
EWC ($\lambda = 10^6$)	24.54% ± 5.55	26.82% ± 3.87	25.25% ± 21.43	37.63% ± 16.46	24.52%
SI ($c = 1$)	1.43% ± 2.86	23.16% ± 2.62	96.33% ± 0.78	96.33% ± 0.78	48.13%
SI ($c = 10^2$)	25.92% ± 15.29	93.64% ± 5.00	94.09% ± 1.55	94.18% ± 1.61	59.38%
SI ($c = 10^4$)	28.42% ± 11.85	95.99% ± 1.68	93.10% ± 1.15	93.15% ± 1.15	60.45%
SI ($c = 10^6$)	56.67% ± 11.27	94.42% ± 3.23	87.66% ± 6.75	88.18% ± 6.68	72.18%
PackNet (20%)	0.22% ± 0.32	37.02% ± 10.78	95.23% ± 1.51	95.23% ± 1.51	46.99%
PackNet (40%)	2.12% ± 1.47	83.97% ± 6.68	95.83% ± 0.66	95.83% ± 0.66	48.27%
PackNet (60%)	9.69% ± 1.65	96.43% ± 0.75	96.39% ± 0.78	96.40% ± 0.79	52.36%
PackNet (80%)	31.39% ± 8.71	97.52% ± 0.55	96.33% ± 0.34	96.38% ± 0.35	63.57%
Random (20%)	0.95% ± 1.30	71.11% ± 14.40	95.97% ± 0.55	95.97% ± 0.55	47.65%
Random (40%)	6.69% ± 3.23	93.64% ± 4.94	95.80% ± 0.90	95.81% ± 0.90	50.61%
Random (60%)	18.59% ± 6.05	96.37% ± 1.52	96.75% ± 0.35	96.76% ± 0.34	57.23%
Random (80%)	44.00% ± 7.49	97.60% ± 0.21	95.65% ± 0.62	95.82% ± 0.66	69.43%
P-Factor $\tau = 0.2$ **(Ours)**	1.33% ± 1.51	52.71% ± 15.35	95.02% ± 1.82	95.02% ± 1.83	47.49%
P-Factor $\tau = 0.4$ **(Ours)**	8.78% ± 4.39	76.40% ± 9.52	96.24% ± 0.78	96.26% ± 0.78	51.76%
P-Factor $\tau = 0.6$ **(Ours)**	31.19% ± 8.03	93.01% ± 4.37	95.70% ± 1.59	95.84% ± 1.51	62.99%
P-Factor $\tau = 0.8$ **(Ours)**	49.57% ± 10.09	96.43% ± 1.24	94.34% ± 0.86	95.37% ± 1.10	71.76%

Table B.4: **Epoch 1: Ablation Studies (Mean ± Std)**.

Method	**Task A (Class-IL)**	**Task A (Task-IL)**	**Task B (Class-IL)**	**Task B (Task-IL)**	**Combined**
Weight Scaling Ablation					
No-Scale $\tau = 0.2$	0.25% ± 0.20	40.90% ± 4.22	95.77% ± 1.20	95.77% ± 1.20	47.22%
No-Scale $\tau = 0.4$	0.47% ± 0.58	47.25% ± 15.59	95.64% ± 0.96	95.64% ± 0.96	47.30%
No-Scale $\tau = 0.6$	8.62% ± 3.59	83.01% ± 6.64	96.36% ± 0.75	96.38% ± 0.76	51.76%
No-Scale $\tau = 0.8$	15.32% ± 8.55	92.01% ± 9.44	96.14% ± 0.46	96.20% ± 0.48	55.08%
Novice Reset Strategy Ablation					
Reset-Zero $\tau = 0.2$	11.23% ± 4.23	20.08% ± 1.40	71.45% ± 15.06	73.54% ± 13.98	40.82%
Reset-Zero $\tau = 0.4$	16.10% ± 8.33	23.75% ± 2.39	54.43% ± 18.27	60.35% ± 15.22	34.91%
Reset-Zero $\tau = 0.6$	23.89% ± 9.07	33.22% ± 6.97	53.64% ± 19.86	66.25% ± 14.41	38.15%
Reset-Zero $\tau = 0.8$	28.49% ± 5.53	84.77% ± 14.76	91.43% ± 2.42	91.93% ± 2.43	59.57%

Table B.5: **Epoch 3: P-Factor vs Baselines (Mean ± Std).** As the exposure to Task B increases to 3 epochs, the structural integrity of the initial Task A engram is severely tested. The following tables demonstrate that random and magnitude-based freezing strategies begin to experience significant plasticity creep. This intermediate regime highlights the necessity of functional utility metrics like the P-Factor, which target structurally critical neurons rather than relying on naive magnitude heuristics.

Method	**Task A (Class-IL)**	**Task A (Task-IL)**	**Task B (Class-IL)**	**Task B (Task-IL)**	**Combined**
Fine-Tuning (Baseline)	0.00% ± 0.00	19.28% ± 0.31	97.51% ± 0.35	97.51% ± 0.35	48.09%
ER (Buffer=200)	90.70% ± 1.23	96.86% ± 1.32	96.89% ± 0.61	97.33% ± 0.65	94.24%
EWC ($\lambda = 10^3$)	0.59% ± 0.93	46.25% ± 18.32	97.27% ± 0.16	97.27% ± 0.16	48.25%
EWC ($\lambda = 10^5$)	15.89% ± 7.48	22.30% ± 3.29	50.16% ± 38.98	61.45% ± 30.71	32.51%
EWC ($\lambda = 10^6$)	23.62% ± 3.63	24.59% ± 2.45	7.79% ± 15.58	22.98% ± 8.38	15.54%
SI ($c = 1$)	7.60% ± 4.22	86.91% ± 9.26	97.18% ± 0.70	97.18% ± 0.70	51.85%
SI ($c = 10^2$)	18.55% ± 8.29	95.65% ± 1.20	95.54% ± 1.25	95.54% ± 1.25	56.49%
SI ($c = 10^4$)	27.30% ± 6.73	95.28% ± 2.69	95.27% ± 0.98	95.30% ± 0.96	60.92%
SI ($c = 10^6$)	54.97% ± 11.02	87.36% ± 8.15	81.26% ± 9.44	81.96% ± 9.21	68.18%
PackNet (20%)	1.02% ± 0.87	78.53% ± 7.76	97.38% ± 0.58	97.38% ± 0.58	48.57%
PackNet (40%)	8.99% ± 1.15	96.49% ± 0.78	97.76% ± 0.33	97.76% ± 0.33	52.79%
PackNet (60%)	16.22% ± 2.37	97.89% ± 0.29	97.75% ± 0.14	97.75% ± 0.15	56.52%
PackNet (80%)	34.27% ± 5.06	98.37% ± 0.18	98.00% ± 0.13	98.05% ± 0.14	65.86%
Random (20%)	3.08% ± 2.92	88.97% ± 4.44	97.30% ± 0.41	97.30% ± 0.42	49.62%
Random (40%)	8.62% ± 3.35	95.16% ± 1.52	97.52% ± 0.61	97.53% ± 0.61	52.40%
Random (60%)	22.97% ± 7.91	97.20% ± 0.55	97.28% ± 0.31	97.30% ± 0.30	59.79%
Random (80%)	42.47% ± 2.95	98.14% ± 0.27	97.08% ± 0.56	97.25% ± 0.56	69.85%
P-Factor $\tau = 0.2$ (Ours)	7.16% ± 6.92	62.39% ± 16.22	97.36% ± 0.29	97.39% ± 0.28	51.58%
P-Factor $\tau = 0.4$ (Ours)	17.36% ± 9.70	81.37% ± 9.73	97.24% ± 0.42	97.29% ± 0.44	56.85%
P-Factor $\tau = 0.6$ (Ours)	47.40% ± 7.12	94.96% ± 1.20	97.06% ± 0.14	97.38% ± 0.31	72.42%
P-Factor $\tau = 0.8$ (Ours)	64.46% ± 8.99	96.89% ± 1.62	94.66% ± 1.92	96.46% ± 0.79	79.88%

Table B.6: **Epoch 3: Ablation Studies (Mean ± Std)**.

Method	**Task A (Class-IL)**	**Task A (Task-IL)**	**Task B (Class-IL)**	**Task B (Task-IL)**	**Combined**
Weight Scaling Ablation					
No-Scale $\tau = 0.2$	0.64% ± 0.45	62.65% ± 10.44	97.14% ± 0.58	97.14% ± 0.58	48.24%
No-Scale $\tau = 0.4$	3.92% ± 2.28	78.09% ± 10.99	97.38% ± 0.29	97.39% ± 0.29	49.99%
No-Scale $\tau = 0.6$	17.59% ± 6.95	94.24% ± 2.83	96.89% ± 0.35	96.92% ± 0.37	56.79%
No-Scale $\tau = 0.8$	30.36% ± 13.34	95.69% ± 2.86	96.57% ± 0.61	96.80% ± 0.54	63.25%
Novice Reset Strategy Ablation					
Reset-Zero $\tau = 0.2$	8.11% ± 7.01	20.04% ± 0.81	88.01% ± 10.27	88.32% ± 10.00	47.37%
Reset-Zero $\tau = 0.4$	11.56% ± 6.45	24.32% ± 3.63	80.44% ± 21.88	84.30% ± 15.32	45.53%
Reset-Zero $\tau = 0.6$	29.67% ± 7.06	56.90% ± 17.06	73.49% ± 22.49	81.03% ± 14.65	51.45%
Reset-Zero $\tau = 0.8$	46.77% ± 15.09	88.40% ± 8.07	92.13% ± 7.49	96.31% ± 0.60	69.51%

Table B.7: **Epoch 5: P-Factor vs Baselines (Mean ± Std).** The results after 5 epochs represent the asymptotic performance of the models under sustained catastrophic interference. By this stage, any temporary 'warm start' advantages have eroded. Gradient-based methods such as EWC and SI collapse, demonstrating their incompatibility with SNN optimization landscapes. In contrast, among the methods tested, structural isolation provides the most effective defense against forgetting in the absence of an episodic replay buffer.

Method	Task A (Class-IL)	Task A (Task-IL)	Task B (Class-IL)	Task B (Task-IL)	Combined
Fine-Tuning (Baseline)	0.00% ± 0.00	20.00% ± 0.65	97.29% ± 0.86	97.29% ± 0.86	48.09%
ER (Buffer=200)	90.93% ± 0.93	96.10% ± 0.77	96.82% ± 0.60	97.59% ± 0.62	94.42%
EWC ($\lambda = 10^3$)	3.71% ± 4.59	25.04% ± 3.03	97.61% ± 0.12	97.68% ± 0.07	50.14%
EWC ($\lambda = 10^5$)	14.29% ± 8.29	23.12% ± 1.58	62.49% ± 35.81	70.24% ± 29.56	38.00%
EWC ($\lambda = 10^6$)	21.17% ± 10.84	28.18% ± 2.82	23.18% ± 38.00	34.64% ± 31.62	22.15%
SI ($c = 1$)	4.69% ± 2.44	83.17% ± 10.21	97.24% ± 0.58	97.24% ± 0.58	50.39%
SI ($c = 10^2$)	9.70% ± 3.75	92.42% ± 3.55	96.23% ± 0.91	96.24% ± 0.92	52.21%
SI ($c = 10^4$)	24.05% ± 9.27	72.67% ± 22.44	88.25% ± 14.23	91.85% ± 7.16	55.41%
SI ($c = 10^6$)	57.65% ± 6.52	82.79% ± 14.38	72.74% ± 20.59	76.79% ± 16.08	64.87%
PackNet (20%)	5.56% ± 2.65	73.33% ± 21.66	94.17% ± 7.89	94.21% ± 7.81	49.25%
PackNet (40%)	12.13% ± 2.54	96.50% ± 0.76	98.00% ± 0.24	98.01% ± 0.24	54.63%
PackNet (60%)	24.40% ± 3.32	97.65% ± 0.60	97.83% ± 0.31	97.86% ± 0.30	60.82%
PackNet (80%)	35.13% ± 4.69	98.21% ± 0.23	97.89% ± 0.45	97.92% ± 0.45	66.50%
Random (20%)	6.88% ± 2.84	92.77% ± 1.57	97.77% ± 0.67	97.78% ± 0.67	51.85%
Random (40%)	17.38% ± 4.49	96.54% ± 0.59	97.91% ± 0.21	97.92% ± 0.21	57.31%
Random (60%)	25.74% ± 8.81	97.26% ± 0.46	97.18% ± 0.49	97.22% ± 0.48	60.94%
Random (80%)	43.58% ± 6.73	98.18% ± 0.28	97.33% ± 0.56	97.48% ± 0.53	70.35%
P-Factor $\tau = 0.2$ (Ours)	8.94% ± 3.34	59.54% ± 14.95	97.52% ± 0.29	97.54% ± 0.26	52.51%
P-Factor $\tau = 0.4$ (Ours)	12.28% ± 2.28	80.41% ± 6.48	97.57% ± 0.31	97.59% ± 0.31	54.51%
P-Factor $\tau = 0.6$ (Ours)	45.77% ± 5.35	90.79% ± 4.70	97.16% ± 0.42	97.55% ± 0.46	71.29%
P-Factor $\tau = 0.8$ (Ours)	60.19% ± 12.51	93.21% ± 4.80	93.66% ± 3.88	97.58% ± 0.28	77.00%

Table B.8: **Epoch 5: Ablation Studies (Mean ± Std)**.

Method	Task A (Class-IL)	Task A (Task-IL)	Task B (Class-IL)	Task B (Task-IL)	Combined
Weight Scaling Ablation					
No-Scale $\tau = 0.2$	2.24% ± 1.55	61.88% ± 17.63	97.54% ± 0.33	97.55% ± 0.33	49.20%
No-Scale $\tau = 0.4$	7.92% ± 3.35	80.02% ± 7.62	97.92% ± 0.23	97.92% ± 0.24	52.47%
No-Scale $\tau = 0.6$	24.24% ± 7.09	92.01% ± 5.23	97.47% ± 0.38	97.53% ± 0.33	60.54%
No-Scale $\tau = 0.8$	30.26% ± 6.99	92.73% ± 3.39	96.57% ± 0.87	97.09% ± 0.53	63.27%
Novice Reset Strategy Ablation					
Reset-Zero $\tau = 0.2$	12.65% ± 2.02	19.33% ± 0.24	87.29% ± 7.88	88.30% ± 8.43	49.35%
Reset-Zero $\tau = 0.4$	13.40% ± 4.85	22.91% ± 2.66	73.09% ± 18.49	73.56% ± 18.46	42.54%
Reset-Zero $\tau = 0.6$	33.94% ± 3.57	62.26% ± 21.65	66.46% ± 26.39	74.11% ± 22.89	49.92%
Reset-Zero $\tau = 0.8$	54.68% ± 8.06	76.75% ± 16.93	72.21% ± 22.38	80.53% ± 15.58	63.68%

Table B.9: **P-Factor Performance at Epochs 2 and 4 ($\tau = 0.8$).** To further characterize the stability-plasticity trade-off, we evaluate the P-Factor mechanism ($\tau = 0.8$) at 2 and 4 epochs per task. This demonstrates the method's performance under these intermediate adaptation durations.

Epochs	Task A (Class-IL)	Task A (Task-IL)	Task B (Class-IL)	Task B (Task-IL)	Combined
Epoch 2	58.53% ± 21.16	94.74% ± 5.08	93.43% ± 3.41	96.22% ± 0.93	76.23%
Epoch 4	56.66% ± 13.52	95.92% ± 1.77	94.56% ± 2.79	96.72% ± 0.77	75.80%

Table B.10: **Epoch 3: N-MNIST Baselines Comparison (Mean ± Std).** To evaluate the robustness of our P-Factor consolidation mechanism on spatio-temporal data, we extend our analysis to the neuromorphic N-MNIST dataset. Unlike static images converted to Poisson spike trains, N-MNIST features asynchronous event streams with inherent temporal dynamics. The table summarizes all baseline performances, revealing that surrogate gradients fail on event streams (EWC/SI) and that structural isolation is uniquely vulnerable to noise saturation at high capacities.

Method	**Task A (Class-IL)**	**Task A (Task-IL)**	**Task B (Class-IL)**	**Task B (Task-IL)**	**Combined**
Fine-Tuning (Baseline)	0.00% ± 0.01	18.97% ± 0.46	98.22% ± 0.33	98.22% ± 0.33	48.41%
ER (Buffer=200)	91.95% ± 1.49	98.32% ± 0.30	98.04% ± 0.27	98.38% ± 0.11	95.66%
EWC ($\lambda = 10^3$)	1.79% ± 2.20	48.56% ± 21.02	98.27% ± 0.22	98.27% ± 0.22	49.41%
EWC ($\lambda = 10^5$)	4.04% ± 2.41	61.09% ± 15.72	97.89% ± 0.67	97.90% ± 0.66	50.44%
EWC ($\lambda = 10^6$)	6.32% ± 4.10	66.48% ± 15.47	98.26% ± 0.24	98.27% ± 0.25	51.88%
SI ($c = 1$)	6.27% ± 3.29	78.10% ± 11.31	97.80% ± 0.66	97.80% ± 0.66	51.46%
SI ($c = 10^2$)	7.60% ± 1.86	90.21% ± 3.66	97.53% ± 0.91	97.53% ± 0.90	51.98%
SI ($c = 10^4$)	11.92% ± 4.58	97.19% ± 1.22	97.16% ± 0.58	97.16% ± 0.58	53.95%
SI ($c = 10^6$)	34.74% ± 7.12	98.85% ± 0.39	94.69% ± 1.21	95.39% ± 0.76	64.39%
PackNet (20%)	0.69% ± 0.70	67.72% ± 6.28	98.31% ± 0.37	98.31% ± 0.37	48.78%
PackNet (40%)	2.18% ± 2.41	92.61% ± 2.17	98.12% ± 0.43	98.12% ± 0.43	49.57%
PackNet (60%)	3.66% ± 3.01	96.61% ± 0.76	98.22% ± 0.41	98.22% ± 0.41	50.36%
PackNet (80%)	6.92% ± 1.58	98.25% ± 0.34	98.25% ± 0.16	98.25% ± 0.16	52.25%
Random (20%)	0.51% ± 0.23	82.93% ± 5.23	98.28% ± 0.33	98.28% ± 0.33	48.76%
Random (40%)	0.78% ± 0.29	91.79% ± 4.91	98.37% ± 0.38	98.37% ± 0.38	48.88%
Random (60%)	2.20% ± 1.35	96.73% ± 1.06	98.59% ± 0.09	98.59% ± 0.09	49.62%
Random (80%)	6.12% ± 2.61	98.09% ± 0.30	98.26% ± 0.11	98.26% ± 0.12	51.65%
P-Factor $\tau = 0.2$ (Ours)	0.06% ± 0.11	80.46% ± 3.12	97.86% ± 0.55	97.86% ± 0.55	48.33%
P-Factor $\tau = 0.4$ (Ours)	0.04% ± 0.02	79.41% ± 4.94	98.36% ± 0.19	98.36% ± 0.19	48.52%
P-Factor $\tau = 0.6$ (Ours)	0.76% ± 0.72	93.95% ± 2.35	98.34% ± 0.08	98.34% ± 0.08	48.90%
P-Factor $\tau = 0.8$ (Ours)	4.73% ± 4.40	97.61% ± 0.89	98.17% ± 0.16	98.17% ± 0.16	50.85%

Table B.11: **5-Split Fashion-MNIST Epoch 3: Experience Replay Progression over Time.** The following tables detail the progression of Experience Replay and Fine-Tuning baselines on the 5-Split Fashion-MNIST benchmark. The P-Factor progression matrix and its detailed analysis are presented in Section 6.

	After Task 1		**After Task 2**		**After Task 3**		**After Task 4**		**After Task 5**	
Eval Task	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**
Task 1	97.3% ± 3.1	98.8% ± 0.1	80.7% ± 15.0	88.6% ± 18.9	75.3% ± 18.9	79.5% ± 22.0	78.0% ± 14.2	96.9% ± 1.1	88.8% ± 5.8	96.2% ± 1.5
Task 2	–	–	76.3% ± 36.8	87.2% ± 16.8	23.8% ± 18.4	76.0% ± 18.9	28.3% ± 19.0	83.2% ± 9.2	55.7% ± 17.5	88.6% ± 9.3
Task 3	–	–	–	–	77.7% ± 22.7	99.1% ± 1.1	49.7% ± 15.5	96.7% ± 4.1	46.4% ± 17.4	95.1% ± 5.8
Task 4	–	–	–	–	–	–	74.5% ± 19.7	99.8% ± 0.1	31.7% ± 17.2	86.6% ± 19.3
Task 5	–	–	–	–	–	–	–	–	93.3% ± 3.6	99.4% ± 0.5

Table B.12: **5-Split Fashion-MNIST Epoch 3: Fine-Tuning (Baseline) Progression over Time**.

	After Task 1		**After Task 2**		**After Task 3**		**After Task 4**		**After Task 5**	
Eval Task	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**
Task 1	97.3% ± 3.1	98.8% ± 0.1	0.0% ± 0.0	50.4% ± 0.6	0.0% ± 0.0	51.5% ± 2.1	0.0% ± 0.0	57.5% ± 13.6	0.0% ± 0.0	50.4% ± 0.6
Task 2	–	–	52.0% ± 0.7	52.0% ± 0.7	37.9% ± 19.3	50.6% ± 0.7	0.0% ± 0.0	57.8% ± 9.9	0.0% ± 0.0	52.4% ± 3.6
Task 3	–	–	–	–	60.9% ± 19.5	88.0% ± 18.9	12.6% ± 19.0	46.4% ± 16.8	19.5% ± 23.9	54.6% ± 8.9
Task 4	–	–	–	–	–	–	42.0% ± 42.3	58.9% ± 45.0	0.0% ± 0.1	54.3% ± 12.5
Task 5	–	–	–	–	–	–	–	–	30.6% ± 24.9	50.8% ± 0.7

Table B.13: **5-Split Fashion-MNIST Epoch 3: PackNet Progression over Time**.

	After Task 1		After Task 2		After Task 3		After Task 4		After Task 5	
Eval Task	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**
Task 1	98.0% ± 0.7	98.0% ± 0.7	0.0% ± 0.0	90.2% ± 5.1	0.0% ± 0.0	92.2% ± 5.8	0.0% ± 0.0	87.6% ± 11.5	0.0% ± 0.0	90.1% ± 10.6
Task 2	–	–	60.0% ± 5.1	60.0% ± 5.1	61.3% ± 13.0	73.9% ± 15.2	45.5% ± 16.7	71.5% ± 13.9	28.2% ± 17.3	71.3% ± 15.5
Task 3	–	–	–	–	52.0% ± 4.1	99.0% ± 0.9	11.0% ± 7.0	94.1% ± 9.6	9.3% ± 4.5	93.2% ± 8.1
Task 4	–	–	–	–	–	–	60.6% ± 6.2	99.8% ± 0.2	31.4% ± 24.6	99.8% ± 0.2
Task 5	–	–	–	–	–	–	–	–	42.4% ± 46.9	80.1% ± 24.0

Table B.14: **5-Split Fashion-MNIST Epoch 3: Random Freezing Progression over Time**.

	After Task 1		After Task 2		After Task 3		After Task 4		After Task 5	
Eval Task	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**
Task 1	97.3% ± 3.1	98.8% ± 0.1	0.0% ± 0.0	90.8% ± 8.3	0.0% ± 0.0	88.0% ± 16.5	0.0% ± 0.0	90.8% ± 9.9	0.0% ± 0.0	93.2% ± 6.3
Task 2	–	–	87.0% ± 12.1	87.0% ± 12.1	13.5% ± 27.1	77.3% ± 14.5	11.3% ± 22.7	66.5% ± 15.4	12.1% ± 24.1	78.3% ± 16.0
Task 3	–	–	–	–	85.9% ± 28.0	94.0% ± 11.9	37.4% ± 14.2	93.6% ± 5.8	33.9% ± 11.1	85.1% ± 17.2
Task 4	–	–	–	–	–	–	51.9% ± 25.9	87.4% ± 18.3	27.6% ± 19.4	86.0% ± 18.9
Task 5	–	–	–	–	–	–	–	–	15.6% ± 15.5	50.5% ± 0.6

Table B.15: **5-Split Fashion-MNIST Epoch 3: Elastic Weight Consolidation (EWC) Progression over Time** ($\lambda = 10^5$).

	After Task 1		After Task 2		After Task 3		After Task 4		After Task 5	
Eval Task	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**
Task 1	97.3% ± 3.1	98.8% ± 0.1	6.5% ± 5.0	92.2% ± 7.0	2.3% ± 2.9	91.6% ± 9.0	1.4% ± 1.8	94.0% ± 3.2	8.8% ± 9.5	93.2% ± 2.8
Task 2	–	–	94.4% ± 1.6	94.5% ± 1.8	22.5% ± 37.1	79.6% ± 15.3	19.2% ± 38.3	90.0% ± 7.9	8.9% ± 17.9	77.6% ± 17.3
Task 3	–	–	–	–	75.4% ± 38.7	89.7% ± 20.3	0.1% ± 0.2	69.4% ± 19.4	0.1% ± 0.1	55.8% ± 11.9
Task 4	–	–	–	–	–	–	87.3% ± 25.1	98.4% ± 2.9	12.8% ± 24.7	75.3% ± 14.9
Task 5	–	–	–	–	–	–	–	–	72.7% ± 35.0	78.9% ± 25.3

Table B.16: **5-Split Fashion-MNIST Epoch 3: Synaptic Intelligence (SI) Progression over Time** ($c = 10^6$).

	After Task 1		After Task 2		After Task 3		After Task 4		After Task 5	
Eval Task	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**
Task 1	97.3% ± 3.1	98.8% ± 0.1	32.9% ± 6.7	98.8% ± 0.1	19.8% ± 9.8	98.7% ± 0.3	15.8% ± 7.1	98.7% ± 0.3	15.6% ± 7.8	98.7% ± 0.4
Task 2	–	–	92.7% ± 4.7	93.0% ± 4.6	3.9% ± 7.2	91.4% ± 5.8	5.0% ± 9.9	92.0% ± 5.2	5.4% ± 10.6	90.8% ± 6.6
Task 3	–	–	–	–	97.0% ± 4.9	99.5% ± 0.4	8.2% ± 11.0	99.1% ± 0.8	5.5% ± 9.9	99.2% ± 1.1
Task 4	–	–	–	–	–	–	92.5% ± 14.5	99.8% ± 0.1	13.9% ± 5.0	99.8% ± 0.3
Task 5	–	–	–	–	–	–	–	–	75.8% ± 27.8	79.9% ± 24.0

Table B.17: **5-Split Fashion-MNIST Epoch 3: P-Factor (Ours) Progression over Time**.

	After Task 1		After Task 2		After Task 3		After Task 4		After Task 5	
Eval Task	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**	**C-IL**	**T-IL**
Task 1	83.1% ± 18.3	88.9% ± 19.5	22.0% ± 19.7	79.9% ± 22.0	24.4% ± 20.5	78.5% ± 22.6	19.9% ± 16.5	84.2% ± 18.8	20.9% ± 17.0	83.5% ± 19.0
Task 2	–	–	76.6% ± 32.6	84.4% ± 17.1	1.4% ± 2.5	84.5% ± 10.5	18.9% ± 15.8	76.9% ± 18.9	15.3% ± 18.6	76.8% ± 20.4
Task 3	–	–	–	–	91.2% ± 16.8	99.8% ± 0.1	78.4% ± 37.9	96.3% ± 7.0	78.4% ± 39.1	94.5% ± 10.8
Task 4	–	–	–	–	–	–	19.9% ± 39.9	60.0% ± 20.0	9.1% ± 18.1	60.0% ± 19.9
Task 5	–	–	–	–	–	–	–	–	18.7% ± 37.4	60.2% ± 19.8